

Homework Instructions

Submission Deadline: The homework must be submitted within the next two weeks.

Grading Criteria:

- Your assignment will be graded as either **1 (Pass)** or **0 (Fail)**.
- To receive a passing grade, you must solve **75% or more** of the assignment correctly.
- **Feedback Policy:** Detailed feedback on individual solutions will not be provided. After the submission deadline, a complete solution will be published, allowing you to compare your work with the ideal solution. Use this to evaluate your approach and understand any mistakes.

Important:

- Please make sure to submit your work on time. Late submissions will not be accepted.
- Make sure to clearly explain your calculations and reasoning.
- **Solution Steps and Clarifications:** Be prepared! The lecturer may ask any student in your group during class or online to explain the solution steps and reasoning for any problem. Make sure that everyone in your group understands the approach and can explain why the problem was solved in that particular way.
- **Group Contribution:** For each homework problem, list the name(s) of the student(s) from your group who worked on that specific problem. This will help in tracking contributions and ensuring everyone participates.
- **Language:** You may write your answers in **German, English, or a mix of both**. Choose the language you feel most comfortable with.

Homework Problem 1: Multiple choice. (6 Points)

Solve the multiple choice questions (1, 3, 5) of chapter-4 on the companion website of the book.
Show all steps in detail and reason your answer.

Homework Problem 2: True or False. (8 Points)

Solve the True or False questions (1, 2, 3, 5) of chapter-4 on the companion website of the book.
Show all steps in detail and reason your answer.

Homework Problem 3: Longest Prefix Matching. (16 Points)

Consider a datagram network using 8-bit host addresses. Suppose a router uses longest-prefix matching, and has the following forwarding table:

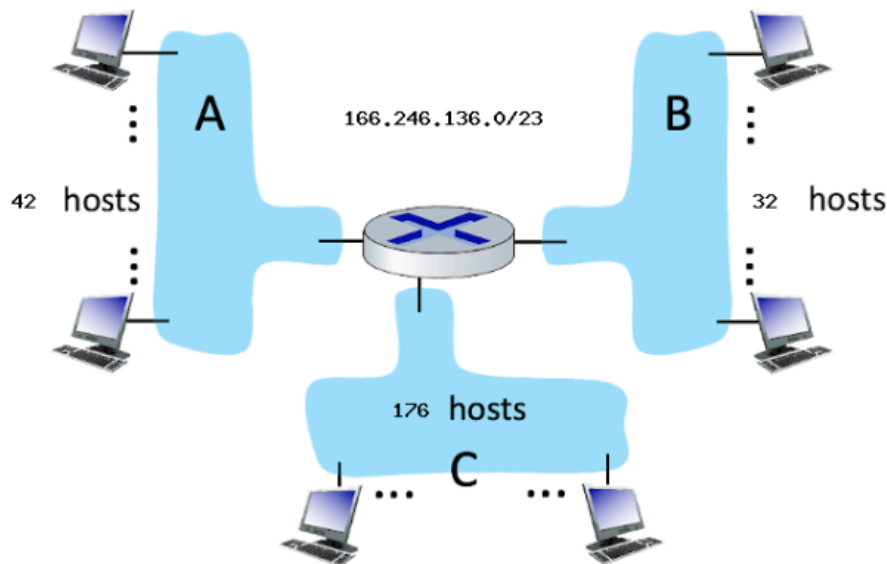
Prefix Match	Interface
00	1
10	2
001	3
101	4
111	5
otherwise	6

To which interface will this datagram be forwarded using longest-prefix matching? Suppose the datagram arrives at the router, with destination addresses:

1. 01000101
2. 01011010
3. 01101110
4. 10101001
5. 10100100
6. 11101101
7. 10100111
8. 11111100

Homework Problem 3: Subnetting. (30 Points)

Consider the router and the three attached subnets below (A, B, and C). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 166.246.136.0/23



Assign subnet addresses to each of the subnets (A, B, and C) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added. Then answer the questions below.

1. Is the address space public or private? Why?
2. How many hosts can there be in this address space?
3. What is the subnet address of subnet A? (CIDR notation)
4. What is the broadcast address of subnet A?
5. What is the starting address of subnet A?
6. What is the ending address of subnet A?
7. What is the subnet address of subnet B? (CIDR notation)
8. What is the broadcast address of subnet B?
9. What is the starting address of subnet B?
10. What is the ending address of subnet B?
11. What is the subnet address of subnet C? (CIDR notation)
12. What is the broadcast address of subnet C?
13. What is the starting address of subnet C?
14. What is the ending address of subnet C?

Homework Problem 4: Packet Capture and Analysis Using Wireshark. (40 Points).

Objective: The purpose of this assignment is to give you hands-on experience with network traffic analysis by using Wireshark (<https://www.wireshark.org/>) to capture and analyze DNS requests. By doing so, you will gain insight into how data packets are structured and transmitted over a network.

Instructions:

- Group Work Requirement: This assignment is a group project; however, each group member must complete the packet capture and analysis individually. Each member's findings will be compiled into a final group report.
- Include screenshots from Wireshark that support each summary.
- Wireshark Setup: You may use Wireshark on any operating system (Windows, macOS, or Linux). Ensure you have the appropriate permissions to run Wireshark, especially on Linux systems, where additional setup might be needed.
- Task: Read the paper, run wireshark and answer all questions on the paper: The paper is available on the companion website of the book. Under wireshark labs **choose DNS** v8.1.

Good Luck